CoRoT-2b

R-1773

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_CoRoT-2_130630` · created 2026-08-02T22:22:24+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456473.7976 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.055
Transit depth (Rp/Rs)^2	2.98 +/- 1.89 [%]
Orbital Inclination (inc)	82.88 +/- 2.1
Airmass coefficient 1 (a1)	1.53 +/- 0.15
Airmass coefficient 2 (a2)	-0.33 +/- 0.12
Scatter in the residuals of the lightcurve fit is	10.03 %
Best Comparison Star	#3 - [286, 254]
Optimal Aperture	2.59
Optimal Annulus	13.13
Transit Duration (day)	0.063 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.173 ± 0.055 vs 0.1667 (deviation 0.1σ)
Duration fit vs published	0.063 d vs 0.095 d (deviation 1.07σ)
Mid-time O–C	-5.8 min
Depth SNR	2.7
Residual scatter	10.03 %

Light curve

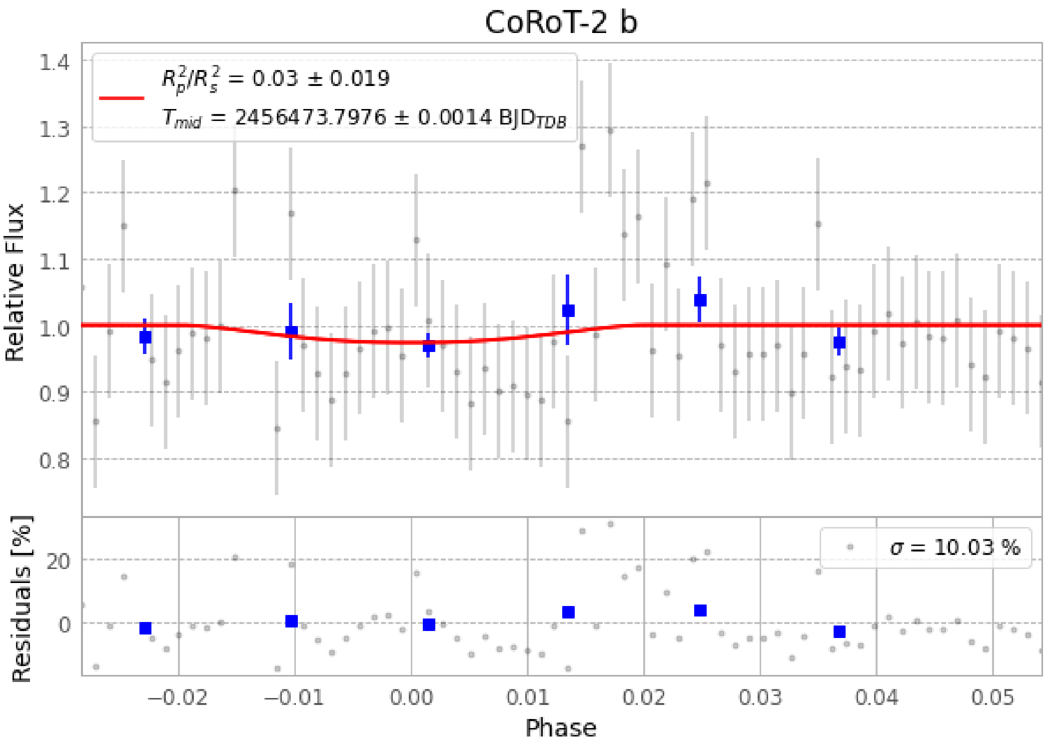

Figure 1 — FinalLightCurve_CoRoT-2 b_2013-06-29.png (SHA-256 05cf2115f43164c9…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [340, 228], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 611db5dc4b5da21b…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

70 FITS frames (46 MB), `CoRoT-2130630054812.FITS` → `CoRoT-2130630091512.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-130630.log (a8c87d755a04…), FinalLightCurve_CoRoT-2 b_2013-06-29.png (05cf2115f431…), FinalLightCurve_CoRoT-2 b_2013-06-29.csv (1e2eec11c358…)

Compute resources

Wall time 175.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)